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Ashim Paudel

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EDUCATION

MS in Computer Science | Louisiana State University | Expected Graduation: December 2026
Concentration: Machine Learning | Selected Courses: Deep Learning, Neural Network, Computer Vision

BS in Computer Science | Mississippi State University | Graduated: May 2022
Major GPA: 3.56 | Selected Courses: DBMS, Programming Languages, AI Robotics, Algorithms, Microprocessors

TECHNICAL SKILLS

Programming: C++, Python, Swift, MySQL, MATLAB, JavaScript
Machine Learning: OpenCV, NumPy, TensorFlow, Scikit-learn
Framework/Library: Boost, ROS, Flask, ARKit, Stereokit
Tools & Utilities: Git, Docker, Azure, AWS, Linux/Unix, Windows

EXPERIENCE

Louisiana State University — Baton Rouge, LA
Graduate Research Assistant — Dec 2024 – Present

- Integrating multi-scale molecular modeling with machine learning to design new soft materials for energy, sustainability, and health applications.
- Combining data science and machine learning to facilitate material design process.

Orbbec — Troy, MI
Research Development Software Engineer — Jul 2022 – Jan 2024

- Developed a real-time image processing pipeline in C++ for converting RGB-D images into stereoscopic view. Integrated Orbbec's proprietary algorithm, leveraging OpenCV and Boost for efficient multi-threaded data handling, and used StereoKit for XR rendering across VR platforms.
- Developed 'The Eye for AIs,' an advanced image processing application that isolates objects, computes volumes, and performs detailed analysis using color and depth images from a depth camera. Leveraged vision language models (VLMs) and large language models (LLMs) to segment and interpret regions of interest from images.
- Led integration of a tailored LLM API into a customer chatbot application. Improved technical response accuracy by 18% by fine-tuning the LLM model using the company software development kit, product specifications, and API documentation.
- Led the design of an iOS prototype to capture and transmit synchronized RGB camera and LiDAR depth sensor data over the network to perform real-time stereoscopic 3D reconstruction. Conducted in-depth research on Apple's LiDAR sensor and depth APIs.
- Collaborated with Nvidia to evaluate the compatibility of their body and hand-tracking technologies with the new Orin and Jetson Nano platforms. Conducted thorough testing of Orbbec SDKs, documenting test cases, outcomes, and errors.

Mississippi State University — Starkville, MS
Undergraduate Research Assistant — Aug 2021 – May 2022

- Assisted in wireless cellular network research, contributing to the setup and execution of experiments. Responsibilities included identifying compatible devices that met the requirements for the testing setup, running test scripts on Ubuntu, and supporting the overall testing process.
- Supported new undergraduates by guiding them through the setup and understanding of srsLTE and ZMQ for research applications. Provided clear explanations of program functions and script operations, enhancing their comprehensive grasp of these research tools.

PROJECTS

- Path Planning using Interactive Application - Built a tool in python that takes in a map image and runs a variety of different path planning algorithms to find the best path between two points while providing an analytical comparison. Tested a variety of algorithms, such as the A* and Dijkstra's algorithm.

AWARDS & ACCOMPLISHMENT

Phi Theta Kappa — 2021